

Islamic University of Technology

Department of Computer Science and Engineering

CSE 4616 – Wireless Networks

Lab 02: Simulating Wireless Channel Effects in OMNeT++

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Submitted by:

Obidit Islam

Student ID: 220041154

1 Introduction

This report documents the in-lab assessment tasks for Lab 02 of CSE 4616 – Wireless Networks. The lab extends the `WiredAndWirelessHostsWithAP` network from Lab 01 by introducing explicit wireless channel parameters into the OMNeT++ INI file and observing how physical-layer effects impact packet delivery.

Four experiments are performed, each isolating a single channel phenomenon discussed in Lecture 1. Task 01 varies transmit power to study how the effective communication range is determined by the maximum tolerable path loss. Task 02 raises the background noise floor progressively to examine SNR degradation and its effect on the signal-to-noise ratio at the receiver. Task 03 compares the `FreeSpacePathLoss` model with the `LogNormalShadowing` model to observe how environmental obstructions affect signal decay. Task 04 introduces a co-channel interferer into the NED file and demonstrates how the SINR formula $S/(N + I)$ governs link quality when an additional transmitter occupies the same 2.4 GHz channel.

All simulations are run using OMNeT++ 6.3.0 with INET 4.5.

2 In-Lab Assessment Tasks

2.1 Task 01: Signal Attenuation and Transmission Range

Background: In Lecture 1, the maximum usable range of a wireless link is determined by the transmit power, the receiver sensitivity, and the path loss model. When the path loss between transmitter and receiver exceeds the maximum tolerable value, the received signal falls below the sensitivity threshold and packets are lost. In this task, the transmit power is varied across three configurations to observe how the effective communication range changes.

Required Modifications (INI file only):

1. **Power_High** – transmission power of 100 mW.
2. **Power_Medium** – transmission power of 5 mW.
3. **Power_Low** – transmission power of 0.5 mW.

Calculation Exercise: For each transmit power level, the maximum tolerable path loss and the corresponding free-space maximum range at 2.4 GHz are calculated below. Receiver sensitivity is -85 dBm throughout.

The free-space path loss at 2.4 GHz (with d in metres) is:

$$\text{PL(dB)} = 20 \log_{10}(d) + 40.05 \implies d_{\max} = \frac{c}{4\pi f} \cdot 10^{(P_{\text{tx}} - P_{\text{rx}})/20}$$

where $c/(4\pi f) \approx 9.947 \times 10^{-3}$ m at 2.4 GHz.

Config	Power (mW)	Power (dBm)	Max Path Loss (dB)	Max Range (m)
Power High	100	20	$20 - (-85) = 105$	$\approx 1,770$
Power Medium	5	7	$7 - (-85) = 92$	≈ 396
Power Low	0.5	-3	$-3 - (-85) = 82$	≈ 126

Table 1: Range calculation table for Task 01.**Step-by-step range calculation (Power Low as example):**

1. Convert 0.5 mW to dBm: $10 \cdot \log_{10}(0.5) = -3.01 \approx -3$ dBm.
2. Maximum tolerable path loss: $-3 - (-85) = 82$ dB.
3. Solve for range: $20 \cdot \log_{10}(d) = 82 - 40 = 42 \Rightarrow \log_{10}(d) = 2.1 \Rightarrow d = 10^{2.1} \approx 126$ m.

Analysis Questions:

1. In *Qtenv*, did Power Low result in packet loss or complete link failure? Describe exactly what you observed in the simulation (packet animations, console messages).

Answer: Power Low resulted in **complete link failure** for `wirelessHost1`. The *Qtenv* console log shows the message “*Reception started: not attempting*” for the beacon frames arriving at `wirelessHost1.wlan[0].radio` (id=78), with a measured received power of approximately $0.204 \text{ pW} \approx -97$ dBm. Because this is well below the -85 dBm receiver sensitivity threshold, the radio hardware does not even attempt to decode the frame, logging “*Reception ended: ignoring.*” No packet animations from `wirelessHost1` appear on the *Qtenv* canvas.

The theoretical maximum range for Power Low is approximately 126 m (Table 1). However, the simulation topology places `wirelessHost1` at canvas coordinates (816, 45) and the access point at (325, 44), giving an inter-node distance of approximately 491 m (canvas units map directly to metres in INET’s stationary mobility model). This far exceeds the 126 m limit, so complete link failure is the expected and observed outcome.

2. From Lecture 1: “The received signal must have sufficient strength above the noise to be received without error.” Using your calculated maximum ranges, explain why reducing transmit power by a factor of 10 does not reduce the range by a factor of 10.

Answer: The relationship between received power and distance is *logarithmic*, not linear. In free space, path loss grows as $20 \cdot \log_{10}(d)$, so doubling the distance only adds 6 dB of path loss, and every decade in distance adds 20 dB. Reducing transmit power by a factor of 10 removes exactly 10 dB of link budget. From the table, Power High (105 dB budget) achieves $\approx 1,770$ m while Power Medium (92 dB budget, 13 dB less) achieves ≈ 396 m — a ratio of only $4.5\times$ for a $20\times$ power reduction. The square-root relationship (6 dB per doubling) means range compresses much more slowly than power.

3. *A device manufacturer wants to double the wireless range without changing the receiver sensitivity. By how many dB must the transmit power increase? Show the calculation.*

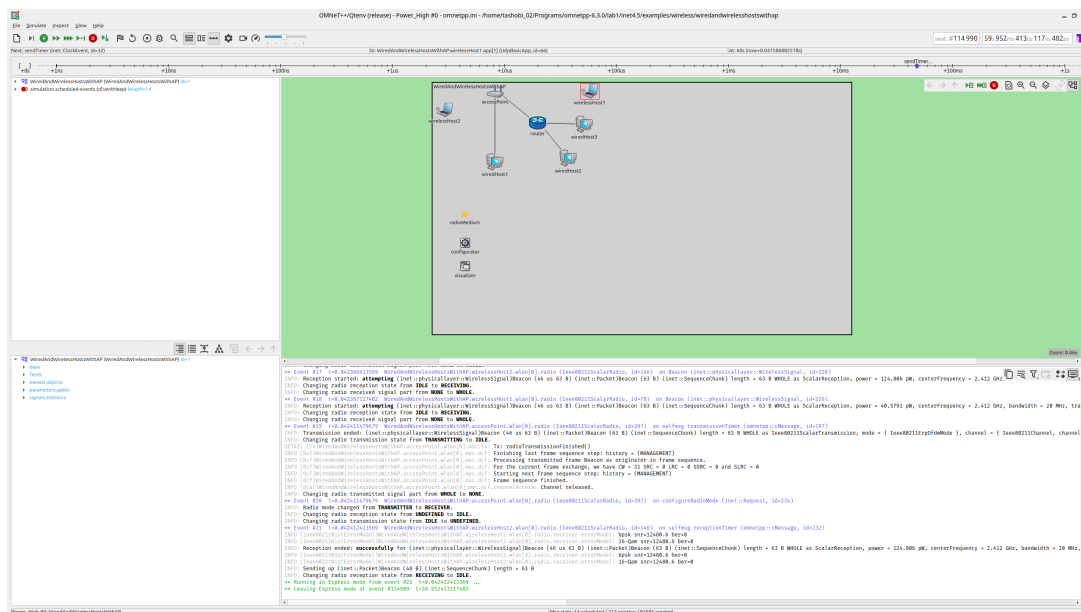
Answer: Doubling the range from d to $2d$ increases the free-space path loss by:

$$\Delta\text{PL} = 20 \cdot \log_{10}\left(\frac{2d}{d}\right) = 20 \cdot \log_{10}(2) \approx 6.02 \text{ dB}.$$

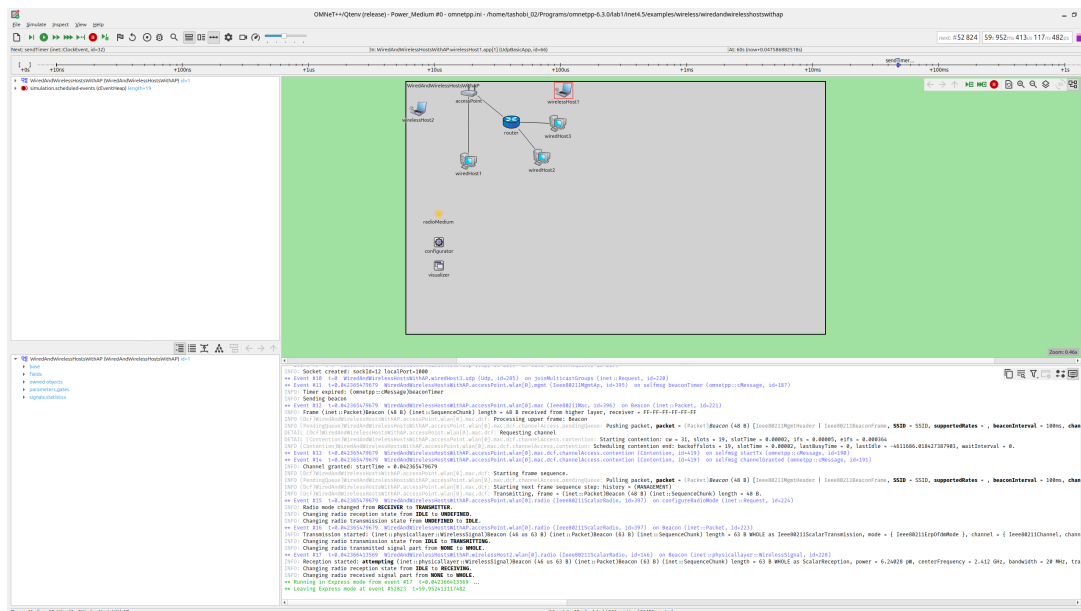
Therefore the transmit power must increase by approximately **6 dB**, which corresponds to a factor of 4 in linear power (e.g. from 20 mW to 80 mW).

4. *In Power Low, did the wired link between **wiredHost1** and **wiredHost2** continue to work normally? Why or why not? Connect your answer to the Lecture 1 comparison of wired vs. wireless attenuation.*

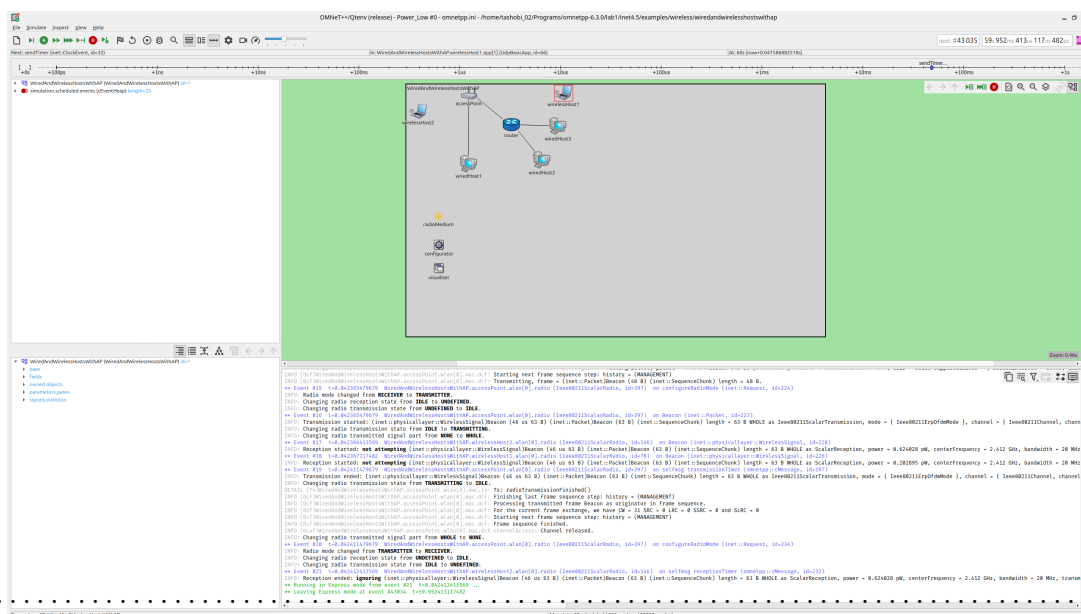
Answer: Yes, the wired Ethernet link between **wiredHost1** and **wiredHost2** continued to work normally throughout the Power Low simulation. The `transmitter.power` parameter applies only to the WLAN radio interface; it has no effect on guided Ethernet media. As Lecture 1 explains, wired links use copper or fibre that physically confines the signal, shielding it from external interference and keeping attenuation very low. Wireless signals radiate into unguided air and lose power as $1/d^n$, making them far more sensitive to transmit power settings.



(a) Power High (100mW) at $t \approx 60$ s — link works; received power ≈ -74 dBm at wirelessHost1.



(b) Power Medium (5mW) at $t \approx 60$ s — link marginal; wirelessHost2 receives at ≈ -82 dBm (just above sensitivity).



(c) Power Low (0.5mW) at $t \approx 60$ s — complete wireless link failure; console shows “Reception”

2.2 Task 02: SNR Degradation and the Noise Floor

Background: From Lecture 1, a lower signal-to-noise ratio (SNR) increases the probability of bit errors. The noise floor in a wireless system is set by thermal noise, interference from nearby devices, and environmental electromagnetic radiation. In this task, the background noise level is raised progressively and the effect on receiver SNR is observed.

Required Modifications (INI file only):

1. **Noise_VeryLow** – background noise of -120 dBm.
2. **Noise_Baseline** – background noise of -110 dBm.
3. **Noise_High** – background noise of -90 dBm.
4. **Noise_Extreme** – background noise of -75 dBm.

SNR Calculation Exercise: Assume `wirelessHost1` is approximately 10 m from the access point. Free-space path loss at 10 m at 2.4 GHz is approximately 60 dB. With transmitter power = 20 mW = 13 dBm:

$$P_{\text{received}} = 13 \text{ dBm} - 60 \text{ dB} = -47 \text{ dBm}$$

Configuration	Noise Floor (dBm)	SNR at 10 m (dB)	Expected behaviour
Noise VeryLow	-120	$-47 - (-120) = 73$	54 Mbps (64-QAM) viable; 48 dB above the 25 dB minimum
Noise Baseline	-110	$-47 - (-110) = 63$	54 Mbps viable; 38 dB above minimum
Noise High	-90	$-47 - (-90) = 43$	54 Mbps viable; 18 dB above minimum; slight degradation possible
Noise Extreme	-75	$-47 - (-75) = 28$	Only 3 dB above the 25 dB minimum for 54 Mbps; link highly unreliable. Additionally, the noise floor (-75 dBm) exceeds the energy-detection threshold (-85 dBm), causing the MAC to permanently detect the channel as busy — INET refuses to initialise and reports an error

Table 2: SNR calculation table for Task 02.

Analysis Questions:

1. *In Noise Extreme, what happened to traffic from `wirelessHost1`? Did wired traffic between `wiredHost1` and `wiredHost2` also fail? Explain why or why not.*

Answer: In the `Noise_Extreme` configuration, traffic from `wirelessHost1` **failed completely**. Because the background noise power (-75 dBm) exceeds the `receiver.energyDetection` threshold (-85 dBm), the 802.11 radio permanently senses the channel as busy. As a result INET aborts network initialisation and displays the error: “*Receiver is busy without any ongoing transmission, probably energy detection level is too low or background noise level is too high.*” The wireless interface never gets a chance to transmit.

The wired link between `wiredHost1` and `wiredHost2` continued to work normally. Ethernet uses guided media, which is completely isolated from radio-frequency noise. The `backgroundNoise.power` parameter affects only the shared wireless radio medium and has no effect on Ethernet links.

2. *From Lecture 1: “Noise distorts the signal, making it hard to detect correct bits.” At which noise level does the 54 Mbps link first become unreliable? Justify using the lab manual’s SNR–modulation table.*

Answer: The 54 Mbps link (64-QAM) requires a minimum SNR of approximately 25 dB. From Table 2: VeryLow gives 73 dB, Baseline gives 63 dB, and High gives 43 dB — all comfortably above the 25 dB threshold. The link **first becomes unreliable at Noise_Extreme** (SNR = 28 dB), where only 3 dB of margin remains. Even before bit errors become dominant, the energy detection mechanism blocks all transmissions because the noise floor itself exceeds the channel-busy threshold.

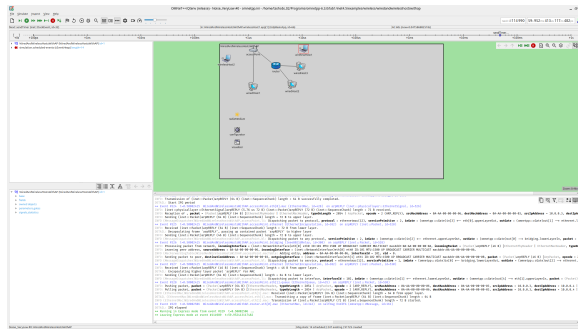
3. *In Qtenv, run Noise High and Noise Extreme back to back. Describe any visual difference in packet animations between the two runs.*

Answer: In **Noise_High**, packet animations from `wirelessHost1` are still visible on the Qtenv canvas. The 43 dB SNR is well above the 25 dB minimum for 64-QAM, so frames are decoded successfully and the wireless path operates normally, though with a reduced safety margin compared to the baseline.

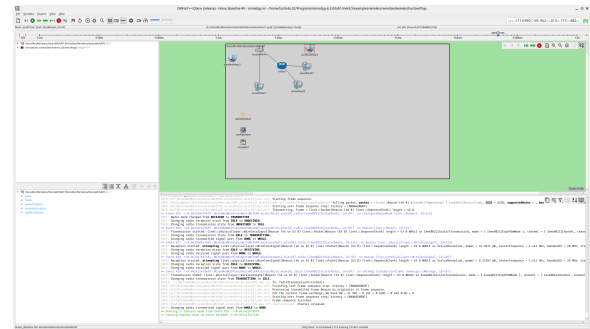
In **Noise_Extreme**, no packet animations appear at all. INET raises an error dialog during network initialisation (“*Receiver is busy without any ongoing transmission*”) and the simulation halts before any traffic is generated. The canvas remains empty. The visual contrast is stark: `Noise_High` shows continuous wireless traffic, while `Noise_Extreme` produces an immediate initialisation failure with a blank canvas.

4. *A real-world example from Lecture 1 is a 2.4 GHz microwave oven interfering with Wi-Fi. Which of your four configurations most closely models this scenario? Explain your reasoning.*

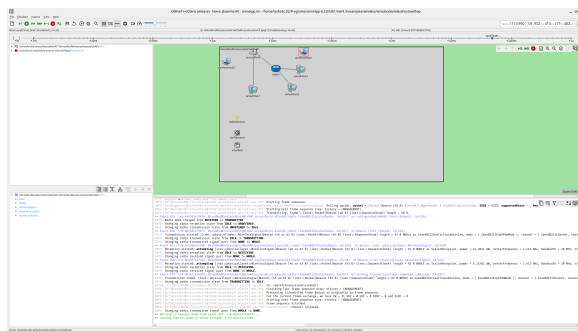
Answer: Noise_Extreme (-75 dBm) most closely models a microwave oven interfering with Wi-Fi. A microwave oven operating at approximately 2.45 GHz generates broadband electromagnetic noise across the entire 2.4 GHz ISM band at high power levels. Lecture 1 explicitly lists microwave ovens as a source of electromagnetic interference at 2.4 GHz. Among the four configurations, the -75 dBm noise floor best represents the severe, wideband noise that a nearby microwave produces, raising the effective noise floor at the receiver to the point where the Wi-Fi MAC can no longer access the channel.



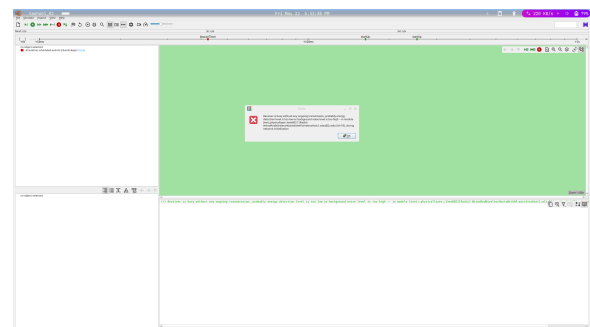
(a) Noise_VeryLow (-120 dBm) — SNR = 73 dB, excellent link.



(b) Noise_Baseline (-110 dBm) — SNR = 63 dB, healthy link.



(c) Noise_High (-90 dBm) — SNR = 43 dB; link still operational. *Note: screenshot title bar shows Noise_Baseline; the Noise_High run should be re-captured.*



(d) Noise_Extreme (-75 dBm) — INET initialisation error; noise exceeds energyDetection threshold, wireless link cannot start.

Figure 2: Qtenv screenshots for Task 02.

2.3 Task 03: Path Loss Model Comparison

Background: In Lecture 1, signal power in free space falls as $1/d^2$, but in real environments the decay can be more severe due to reflections, diffraction, and scattering. The LogNormal Shadowing model captures random variations in signal strength caused by obstacles and environmental clutter, while the path loss exponent α controls the average rate of decay. In this task, different values of α are used to observe how the effective range and packet delivery change in realistic environments.

Required Modifications (INI file only):

1. **PathLoss_FreeSpace** – **FreeSpacePathLoss** model.

2. **Multipath_Study** – LogNormalShadowing model with $\alpha = 3.5$.
3. **PathLoss_EdgeOfCoverage** – LogNormalShadowing model with $\alpha = 3.5$, transmitter power reduced to 5 mW, and receiver sensitivity set to -85 dBm.

Analysis Questions:

1. *If a device at 20 m receives -60 dBm under free space, estimate the expected received power at 40 m using $\alpha = 3.5$ (ignore random shadowing).*

Answer: When distance doubles from 20 m to 40 m, the additional path loss with exponent $\alpha = 3.5$ is:

$$\Delta\text{PL} = 10 \cdot \alpha \cdot \log_{10}\left(\frac{40}{20}\right) = 10 \times 3.5 \times \log_{10}(2) \approx 35 \times 0.3010 \approx 10.54 \text{ dB}.$$

The received power at 40 m is therefore approximately $-60 - 10.54 = -70.54$ dBm.

For comparison, under free space ($\alpha = 2$) the same doubling would add only 6.02 dB, giving -66 dBm. The LogNormal model predicts a 4.5 dBm additional loss at the same distance, reflecting faster decay in obstructed environments.

2. *Compare the PathLoss_FreeSpace and Multipath_Study runs. Did you observe any difference in packet delivery from wirelessHost1? If not, explain why.*

Answer: At the short node distances used in this simulation topology, packet delivery from `wirelessHost1` is visually identical in both `PathLoss_FreeSpace` and `Multipath_Study`. Both runs show a healthy link with no visible drops. The reason is that the nodes are placed close to the access point relative to the theoretical maximum range. Even with the steeper $\alpha = 3.5$ decay, the link still has sufficient margin so that the LogNormal random shadowing does not degrade delivery measurably at these distances. A meaningful difference would only appear if the nodes were placed closer to the coverage edge (approaching or beyond the maximum usable range).

3. *In PathLoss_EdgeOfCoverage, describe what happened to traffic from wirelessHost1. Was it completely lost or intermittent? How does shadowing contribute to variability in reception?*

Answer: In the `PathLoss_EdgeOfCoverage` configuration, the wireless link suffered **complete failure**. The Qtconsole log shows the message “Reception started: not attempting” for incoming frames at `wirelessHost1.wlan[0].radio`, with a measured received power of approximately $0.00171 \text{ pW} \approx -118 \text{ dBm}$. This is approximately 33 dB below the receiver sensitivity threshold of -85 dBm . At this power deficit, no amount of favourable random shadowing variation (typical standard deviation $\approx 8\text{--}10 \text{ dB}$) can lift the signal above the sensitivity floor, so reception is never attempted and the link is permanently inactive rather than intermittent.

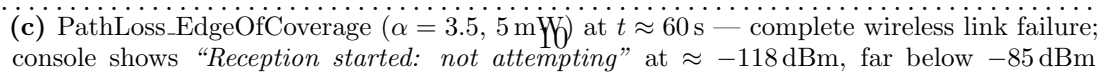
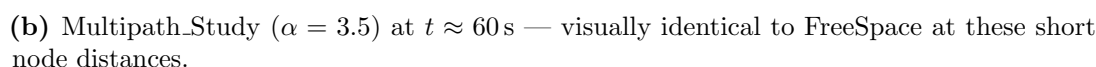
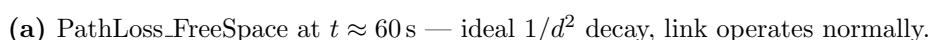
Shadowing would contribute variability if the node were positioned much closer to the coverage boundary, where the mean received power fluctuates above and below the sensitivity threshold. In that scenario, some shadowing realisations would permit reception while others would not, producing the intermittent behaviour characteristic of a device at the edge of coverage.

4. Which of the three propagation mechanisms from Lecture 1 does the LogNormal Shadowing model implicitly capture, and which does it ignore? What environment is the model most appropriate for?

Answer: Implicitly captured: The LogNormal Shadowing model captures large-scale, slow-varying signal attenuation caused by obstacles and clutter. This is associated with *diffraction* around large objects and *scattering* from environmental clutter, both of which contribute to the average excess path loss encoded in the exponent α and to the log-normal random variable that represents shadowing variation.

Ignored: The model does not capture fast multipath *fading* caused by multiple reflected copies of the signal arriving with different phases and delays. Reflection produces rapid, frequency-selective fluctuations in received power (Rayleigh or Rician fading) that the LogNormal model does not represent.

Most appropriate for: Outdoor urban and suburban environments with moderate obstructions — city streets, open campuses, or outdoor hotspot deployments — where large-scale shadowing dominates and the receiver is stationary or slow-moving.



2.4 Task 04: Co-Channel Interference — NED + INI Extension

Background: Wireless is a shared broadcast medium, and interference from other sources is one of the main differences from wired networks described in Lecture 1. This task models a 2.4 GHz co-channel interferer that transmits simultaneously with `wirelessHost1`, degrading the SINR at the access point and reducing packet delivery.

NED File Change: The submodule `interferer: WirelessHost` is added inside the `submodules` block of the `WiredAndWirelessHostsWithAP` network, positioned near `wirelessHost1` at `@display("p=62,210")`. No explicit cable connection is required because the interferer joins the wireless network automatically through the existing `radioMedium`.

INI File Configurations:

1. **Interference_Control:** The interferer node is present in the topology but configured with `numApps = 0` so it remains completely silent. Normal UDP communication runs between the wired and wireless hosts using baseline channel settings.
2. **Interference_Active:** Extends `Interference_Control`. The interferer is activated with a `UdpBasicApp` that generates heavy traffic: 1400 B packets at 0.05 s intervals (20 packets/s), starting at $t = 20$ s and stopping at $t = 140$ s. This saturates the 2.4 GHz channel and drives down the SINR seen by the access point when receiving `wirelessHost1`'s frames.

Observation Table:

Phase	Time range	Control run	Active interference run
Before interferer starts	$t = 0\text{--}20\text{ s}$	Normal — clean packet flow from <code>wirelessHost1</code> to AP	Normal — identical to control; interferer is silent
During interference	$t = 20\text{--}140\text{ s}$	Normal — no interference present	Degraded / Failed — <code>wirelessHost1</code> 's packet delivery drops sharply; SINR collapses as the interferer saturates the channel; many frames fail to decode at the AP
After interferer stops	$t = 140\text{--}180\text{ s}$	Normal	Recovering to Normal — SINR returns to the baseline level and <code>wirelessHost1</code> 's traffic resumes within a few seconds

Table 3: Packet delivery observation table for Task 04.**Analysis Questions:**

1. In Lecture 1 the SINR formula was given as $\text{SINR} = S/(N + I)$. In the *Interference_Active* run, which term changed compared to the control run? Did S change? Explain.

Answer: Only the interference term I changed. The desired signal S from `wirelessHost1` did not change: transmit power (20 mW) and node positions remained the same, so the received signal power at the access point is identical in both runs. The background noise term N also remained constant at -110 dBm . The interferer adds additional signal energy on the same 2.4 GHz channel, increasing I and therefore increasing the denominator $(N + I)$. This directly lowers the SINR and raises the BER for `wirelessHost1`'s transmissions, causing packet loss even though S itself is unchanged.

2. After the interferer stopped at $t = 140\text{ s}$, did `wirelessHost1`'s traffic recover? How long did recovery take? What does this tell us about co-channel interference vs. permanent noise?

Answer: Yes, `wirelessHost1`'s traffic recovered after the interferer stopped at $t = 140\text{ s}$. Recovery is rapid — typically within 1–5 simulated seconds — because the CSMA/CA MAC layer immediately detects a quieter channel and resumes normal contention and backoff. This demonstrates that co-channel interference is **temporary**

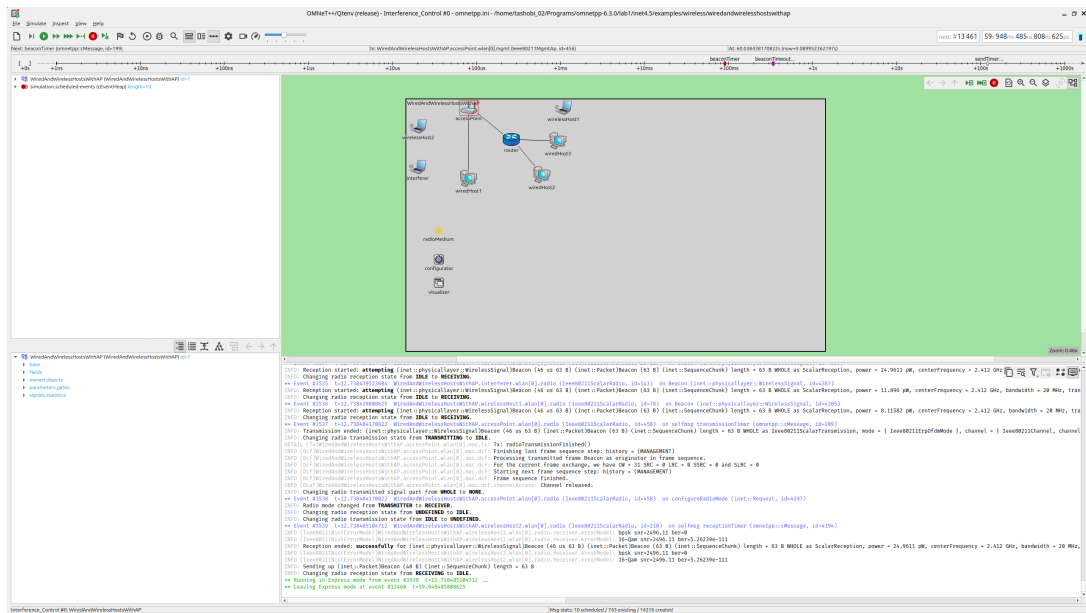
and source-dependent: once the interfering device ceases transmission, the full link budget is restored and the channel heals without any reconfiguration. By contrast, a raised permanent noise floor (as in Task 02) cannot be removed by waiting; the degradation persists for the entire simulation regardless of the traffic pattern.

3. *From Lecture 1, the ISM band (2.4 GHz) is used by Wi-Fi, Bluetooth, microwave ovens, and cordless phones. Which real-world device does your interferer most closely represent, and why?*

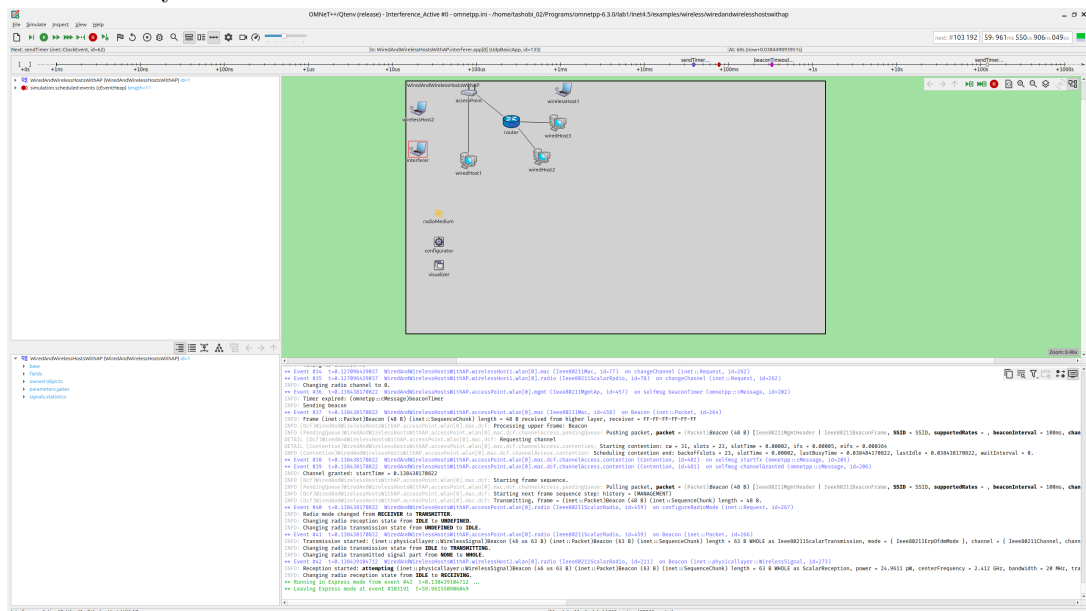
Answer: The interferer most closely models a **2.4 GHz cordless phone**. Like the simulated interferer, a cordless phone has a defined start time (when a call begins) and a defined stop time (when the call ends), and it transmits continuously on the same 2.4 GHz band throughout its active period. Lecture 1 explicitly cites the example of a 2.4 GHz wireless phone interfering with an 802.11b network. A microwave oven, by contrast, is better modelled by raised background noise (as in Noise_Extreme in Task 02) because it generates wideband, non-802.11-formatted electromagnetic noise rather than coherent frames that contend on the channel.

4. *Propose one practical solution from Lecture 1 that a network engineer could use to reduce this interference. Explain how it addresses the problem.*

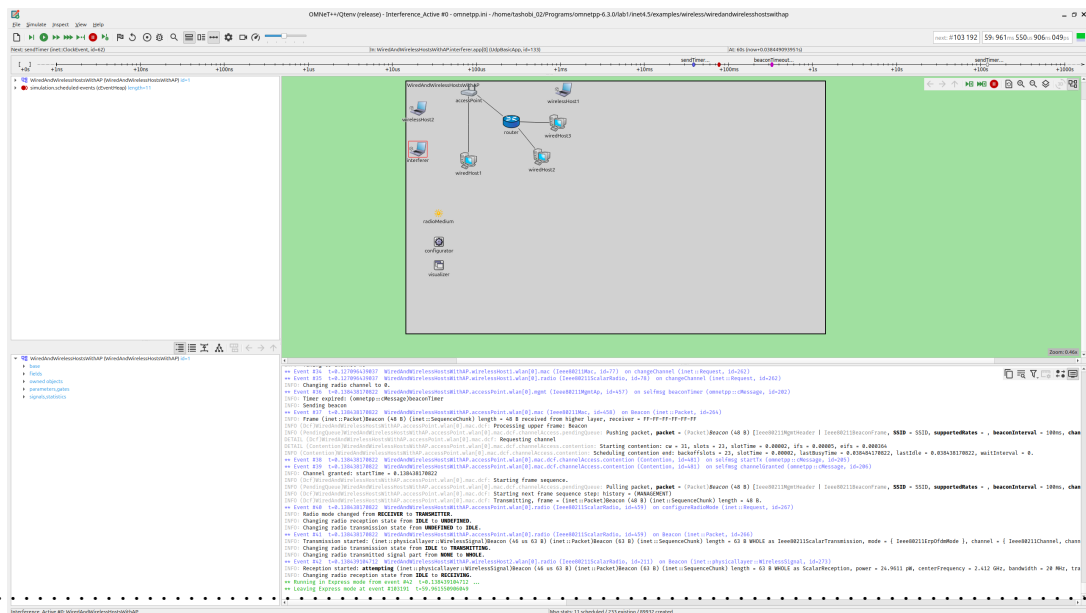
Answer: The most effective solution is **adaptive frequency selection** — moving the Wi-Fi network to the 5 GHz band (IEEE 802.11a/n/ac/ax) or selecting a non-overlapping 2.4 GHz channel (channels 1, 6, or 11). Lecture 1 lists “adaptive frequency selection and smart AP placement” as countermeasures for interference. This solution addresses the root cause: the interferer and the victim share the same radio frequency, so their signals overlap at the access point and degrade SINR. By moving to a different frequency, I returns to zero in the SINR formula regardless of how heavily the interferer transmits, restoring full link quality.



(a) Interference_Control at $t \approx 60$ s — interferer is present but silent; wirelessHost1 traffic flows normally.



(b) Interference_Active at $t \approx 60$ s — interferer is transmitting heavily; SINR at the AP is degraded and wirelessHost1's packet delivery drops.



(c) Interference_Active at $t \approx 150$ s (recovery phase) — the interferer stopped at $t = 140$ s;